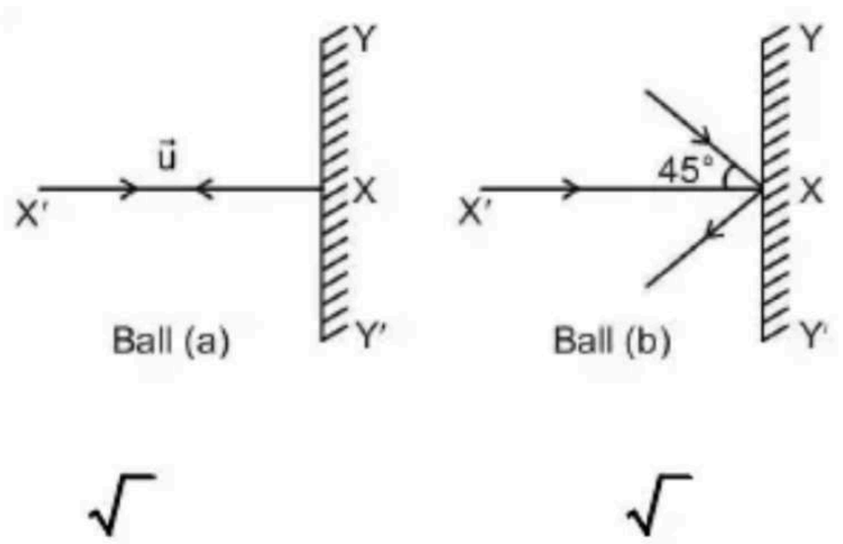
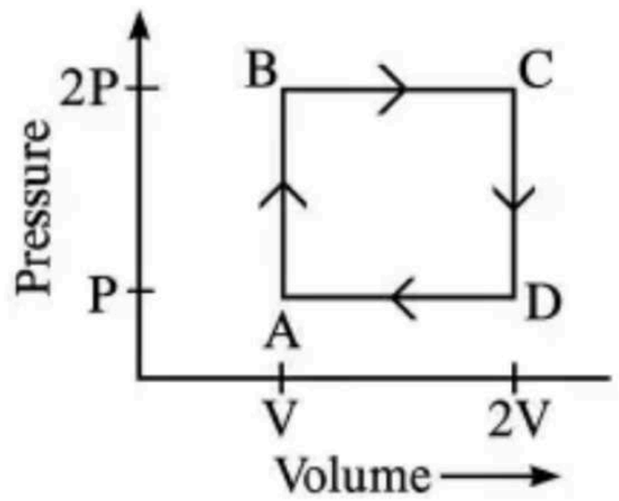
Sri Chaitanya
Educational Institutions



√ √

Sri Chaitanya
Educational Institutions



$\frac{3}{2}$

ω

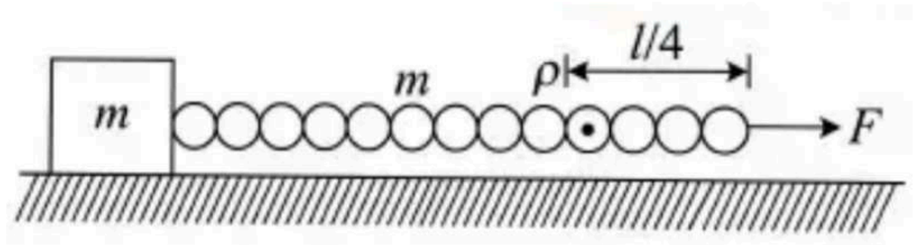
$U \quad x \qquad x \qquad x$

$g \quad g$

$^{2}R \qquad - \quad R$

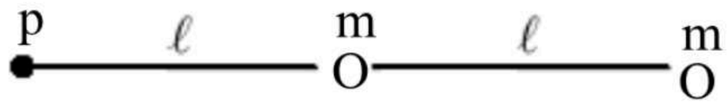
$v \qquad x \qquad \sqrt{\ }$

n



$2^{3} \quad 3^{2} \quad 4 \text{ m}$

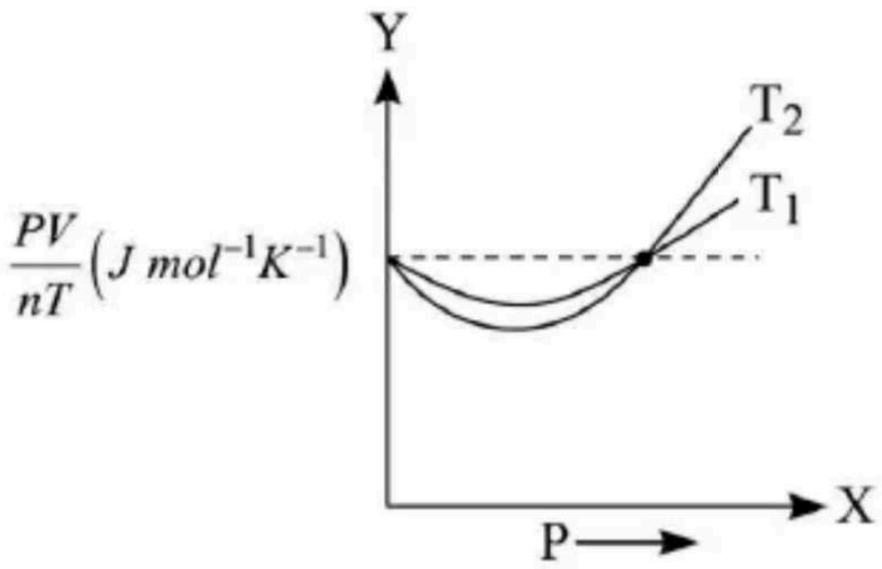
17. Two identical particles, each of mass m, are connected to a string, and the system is rotated in a horizontal plane with p as the centre. The ratio of tension in the two parts of the string is



(1) 1:2 (2) 2:1

(3) 3:2 (4) 1:1

18. The figure below shows the plot of $\frac{PV}{nT}$ versus P for oxygen gas at two different temperatures.



A. The dotted line corresponds to the 'ideal' gas behaviour.

B. $T_1 > T_2$

C. The value of $\frac{PV}{nT}$ at the point where the curves meet on the y-axis is the same for all gases.

Which of the above statements is true?

(1) A only

(2) A and B only

(3) A, B, and C

(4) A and C only

19. An unpolarised light is incident on a plate of refractive index $\sqrt{3}$, and the reflected light is found to be completely plane polarised. The angles of incidence and refraction, respectively, are

(1) 60°, 30°

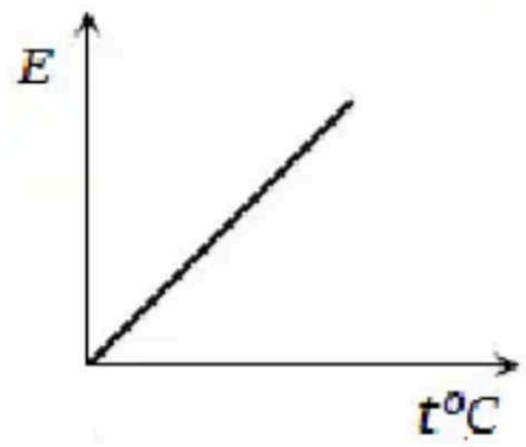
(2) 30°, 60°

(3) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right), 45^0$

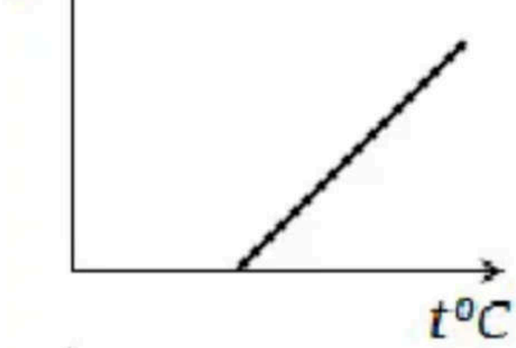
(4) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right), 30^0$

20. The graph that represents the variation of mean kinetic energy of molecules with temperature t°C is

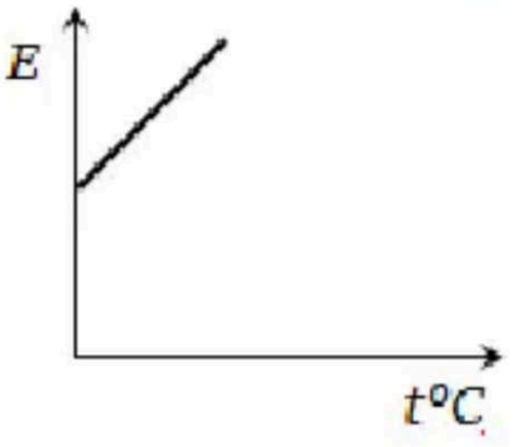
(1)



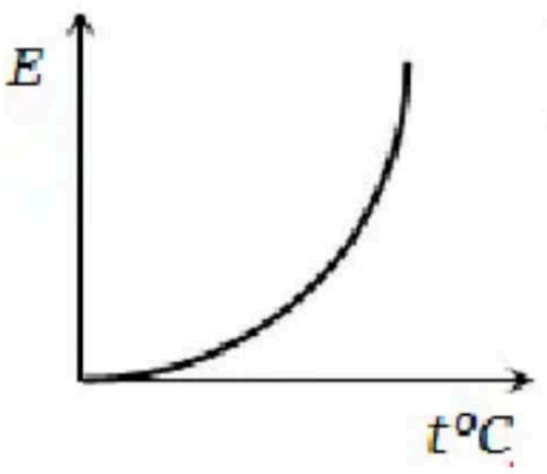
(2)



(3)



(4)



21. A current element of strength $\sqrt{2}\times10^{-4}$ Am is placed at the corner A of a square ABCD of side 10 cm. Magnetic field induction at the point, which is at the diagonally opposite corner of the square due to the element, is (The element lies along the side AB)

(1) 0.5×10^{-9} Wb/m^2

(2) 1.2×10^{-9} Wb/m^2

(3) 2×10^{-9} Wb/m^2

(4) zero

22. Match List I with List II.

	List I (Waves)		**List II (Production)**
A.	Infra-red	I.	Rapid vibration of electrons in aerials
B.	Radio	II.	Electrons in atoms emit photons when they move from a higher to a lower energy level.
C.	Light	III.	Klystron valve
D.	Microwave	IV.	Vibration of atoms and molecules

Choose the correct answer from the options given below.

(1) A-I B-III C-IV D-II

(2) A-IV B-I C-II D-III

(3) A-II B-I C-IV D-III

(4) A-IV B-III C-I D-II

23. The magnetic susceptibility of a paramagnetic substance at –173°C is 1.5×10^{-2}. To have the susceptibility 0.5×10^{-2}, the change in temperature in °C is

(1) 100

(2) 180

(3) 200

(4) 220

24. A series LCR circuit has a resistance 30Ω, an inductance 2.5 H, and a capacitance 5 mF. If the angular frequency of AC fed is 20 rad/s, the power factor is

(1) 0.75 (2) 0.5

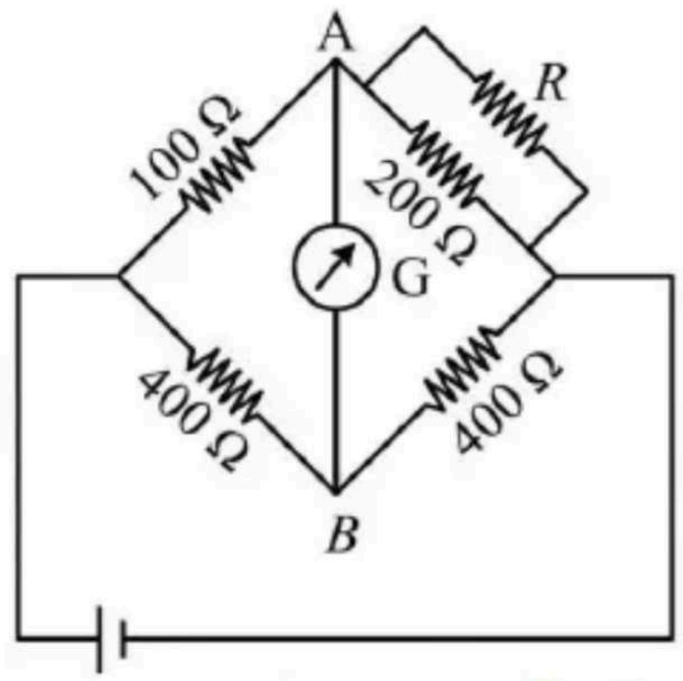
(3) 0.2 (4) 0.6

25. A resistance of 2 Ω is connected across one gap of a meter-bridge (the length of the wire is 100 cm), and an unknown resistance, greater than 2 Ω, is connected across the other gap. When these resistances are interchanged, the balance point shifts by 20 cm. Neglecting any corrections, the unknown resistance is

(1) 3 Ω (2) 4 Ω

(3) 5 Ω (4) 6 Ω

26. The given Wheatstone bridge shows no deflection in the galvanometer joined between the points A and B. If the emf of the battery is 160 V, then the current through the resistance R as shown in the circuit is



(1) 1 A (2) 0.8 A

(3) 0.4 A (4) 0.2 A

27. Choose the INCORRECT statement.

A. The electric lines of force entering into a closed Gaussian surface provide negative flux.

B. A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same.

C. In a uniform electric field, the net flux through a closed Gaussian surface containing no net charge is zero.

D. When an electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

(1) D only (2) C and D only

(3) A and C only (4) B and D only

28. When YDSE is carried out in a liquid, the 5th bright fringe lies at a point on the screen where the 3rd dark fringe was lying when the medium was air. The refractive index of the liquid is

(1) 1.5 (2) 3

(3) 2 (4) 1.2

29. The ionisation potential of H atom is -13.6 eV. Photons of energy 12.75 eV are made to incident on a sample of H atoms at ground state. How many spectral lines are expected in the emitted radiation?

(1) 2 (2) 6

(3) 4 (4) 1

30. A ray of light is incident at an angle of $60°$ on one face of a prism of angle $30°$. The emergent ray of light makes an angle of $30°$ with the incident ray. The angle made by the emergent ray with the second face of the prism will be

(1) $0°$ (2) $90°$

(3) $30°$ (4) $45°$

31. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : A comb run through one's dry hair attracts small bits of paper.

Reason (R) : When a plastic comb rubbed against dry hair gets charged by friction, and when it is brought close to paper, the molecules of paper get polarised by the charged comb.

In light of the above statements, choose the correct answer from the options given below.

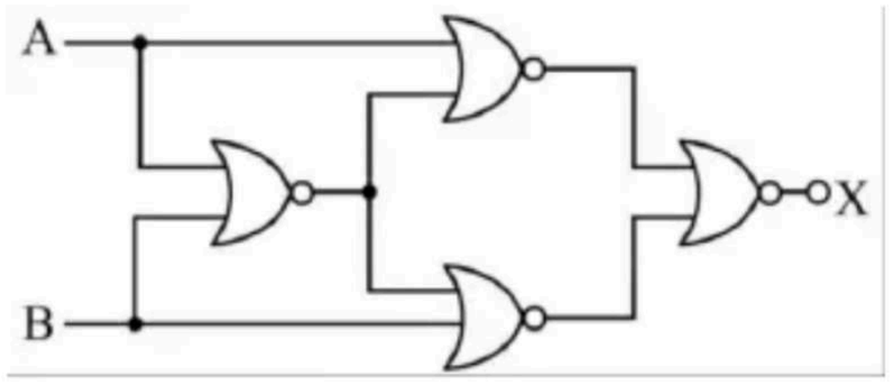
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

32. Select the correct truth table for the system of four NOR gates shown in the figure given below.



(1)

A	B	X
0	0	1
0	1	1
1	0	0
1	1	0

(2)

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

(3)

A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

(4)

A	B	X
0	0	0
0	1	0
1	0	1
1	1	1

33. The Maxwell's equation $\int \vec{B}.\vec{dl} = \mu_o \left[I + \in_0 \frac{d\phi_\in}{dt} \right]$ is a statement of

(1) Faraday's law of induction

(2) Modified Ampere's law

(3) Gauss's law of electricity

(4) Gauss's law of magnetism

34. The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.47 MeV. The energy required to remove a neutron from C^{13} is

(1) 0.21 MeV

(2) 2.52 MeV

(3) 4.95 MeV

(4) 2.75 MeV

35. A closely wound solenoid 80 cm long has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm. If the current carried is 8.0 A, the magnitude of B inside the solenoid near its centre is

(1) 1.5×10^{-2} T, perpendicular to the axis of solenoid

(2) 2.5×10^{-2} T, along the axis of solenoid

(3) 3.5×10^{-2} T, along the axis of solenoid

(4) 1.5×10^{-2} T, along the axis of solenoid

36. A current i flows along the length of an infinitely long, straight, thin-walled pipe. Then the magnetic field

(1) at all points inside the pipe is the same, but not zero

(2) is zero only on the axis of pipe

(3) is different at different points inside the pipe

(4) at any point inside the pipe is zero

37. A lens of power 16 D is used as a simple microscope. In order to obtain maximum magnification, at what distance from the lens should a small object be placed?

(1) 5 cm (2) 10 cm

(3) 16 cm (4) 25 cm

38. A p-n junction diode can be used as a

A. LED

B. rectifier

C. transformer

D. voltage regulator

Choose the correct answer from the options given below.

(1) A and B only

(2) B and C only

(3) A and D only

(4) A, B, and D only

39. Match List I with List II.

List I		List II
A.	Stopping potential	I. $\frac{hc}{eV_0}$
B.	Slope of stopping potential vs frequency graph	II. $\frac{h}{mv}$
C.	de-Broglie wavelength	III. $\frac{h}{e}$
D.	Minimum wavelength of X-rays	IV. $\frac{KE_{max}}{e}$

All the terms have their usual meaning.

Choose the correct answer from the options given below.

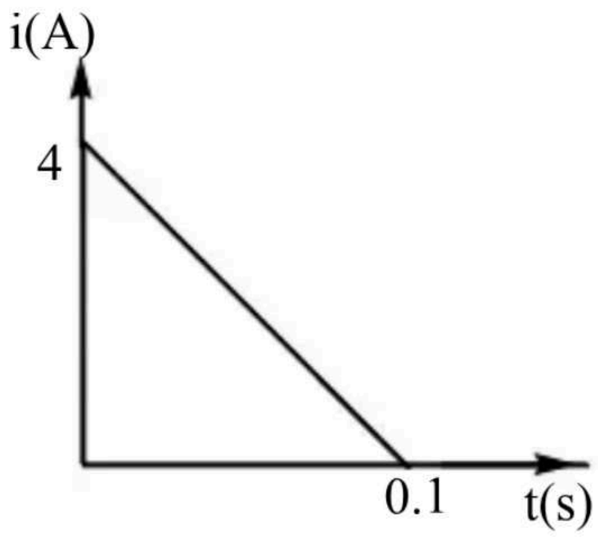
(1) A-IV B-III C-II D-I

(2) A-II B-III C-I D-II

(3) A-IV B-II C-III D-I

(4) A-I B-IV C-III D-II

40. When magnetic flux through a coil changes by 2 Wb, the variation of induced current in the coil with time is as shown in the figure. Then the resistance of the coil will be



(1) 10 Ω

(2) 20 Ω

(3) 5 Ω

(4) 2 Ω

41. The focal lengths of the lenses of an astronomical telescope are 50 cm and 5 cm. The length of the telescope when the image is formed at the least distance of distinct vision is

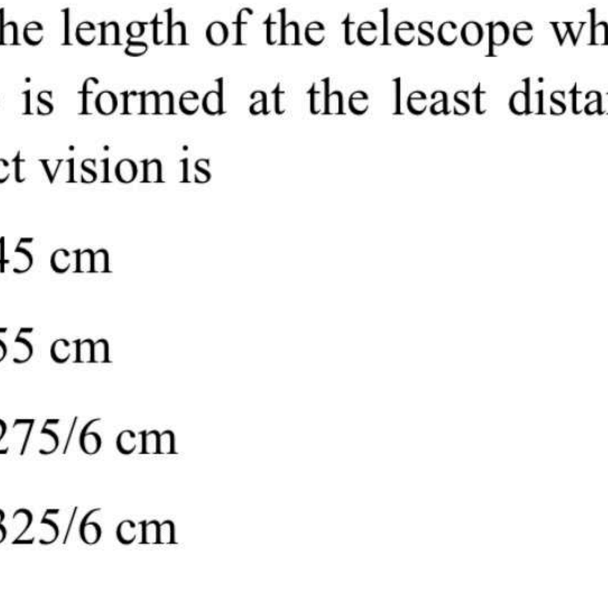
(1) 45 cm

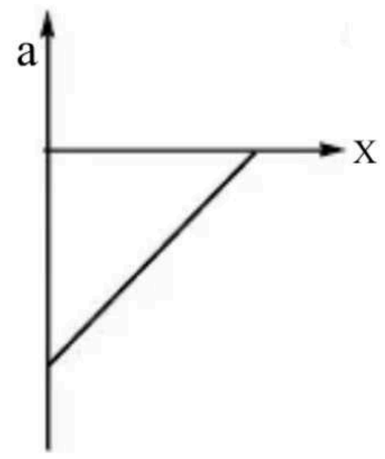
(2) 55 cm

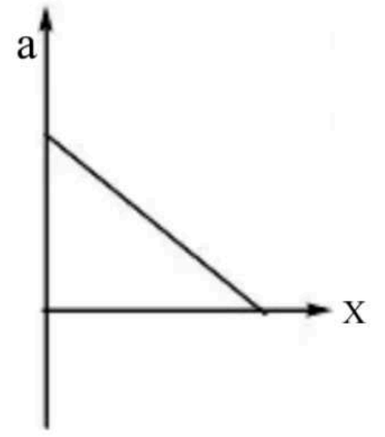
(3) 275/6 cm

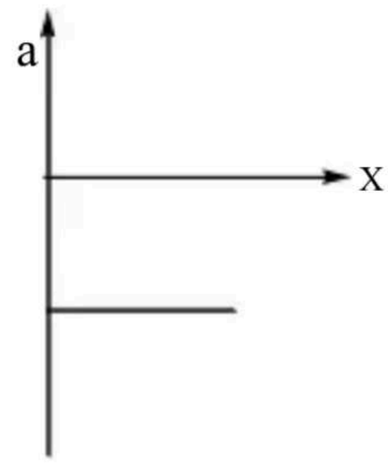
(4) 325/6 cm

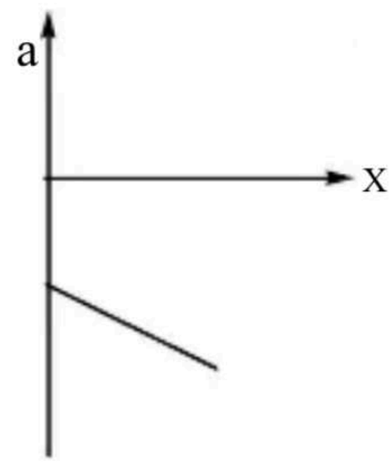
42. The given graph shows the variation of velocity with displacement. Which one of the graphs given below correctly represents the variation of acceleration with displacement?



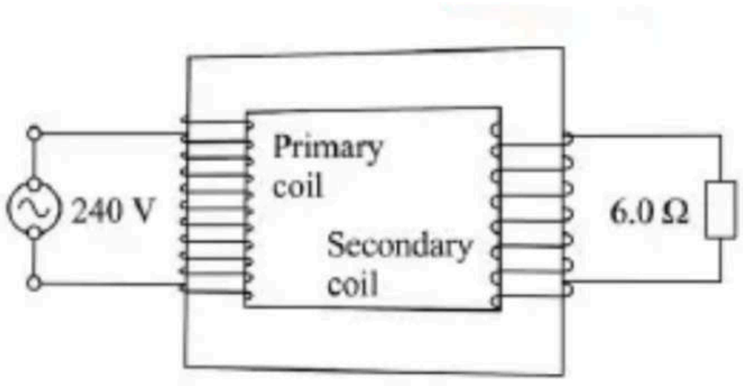
(1)


(2)


(3)


(4)


43. The given figure shows an iron-cored transformer assumed to be 100% efficient. The ratio of the secondary turns to the primary turns is 1:20. A 240 V ac supply is connected to the primary coil, and a 6 Ω resistor is connected to the secondary coil. What is the current in the primary coil?



(1) 0.10 A (2) 0.14 A

(3) 2 A (4) 40 A

44. A spherical conductor of radius 'R' is given a charge 'q'. Find the difference in electric potentials between the following two points:

(i) $R/2$ from centre of sphere

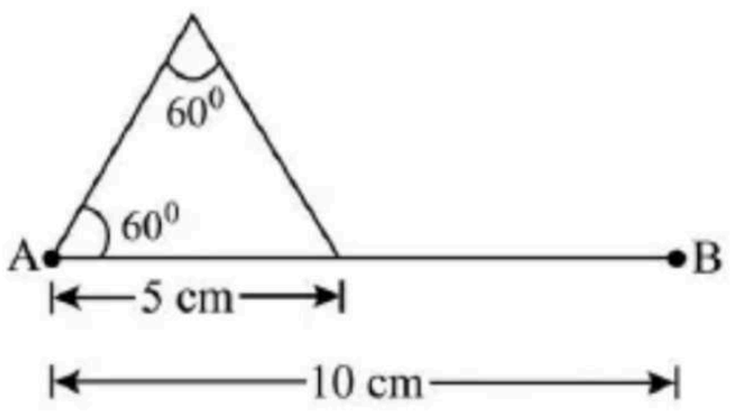
(ii) $R/2$ from the surface outside the sphere

$\left(K = \frac{1}{4\pi\varepsilon_0}\right)$

(1) zero (2) $\frac{Kq}{R}$

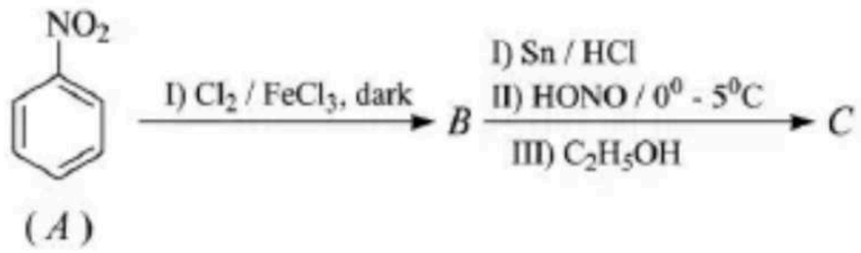
(3) $\frac{2}{3}\frac{Kq}{R}$ (4) $\frac{1}{3}\frac{Kq}{R}$

45. A uniform wire with a resistance of 24 Ω is bent in the following shape. The effective resistance between A and B is



(1) 24 Ω (2) 10 Ω

(3) $\frac{16}{3}\Omega$ (4) 12 Ω

46. 

Most reactive towards acetyl chloride in the presence of anhydrous $AlCl_3$ is

(1) B (2) A

(3) A = B = C (4) C

47. The correct statements among the following are

A. When a system is in equilibrium, the entropy is maximum, and the change in entropy, $\Delta S = 0$.

B. ΔS_{total} can discriminate between isothermal reversible process and isothermal irreversible process.

C. Enthalpy of neutralisation of H_2SO_4 vs NaOH is about -114.6 kJ mol^{-1}.

D. Standard enthalpy of formation of white phosphorous is zero.

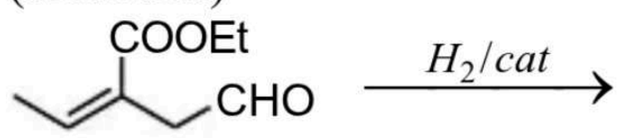
(1) A and C only (2) A, B, C and D

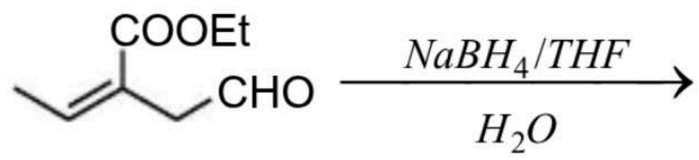
(3) B and D only (4) A, B, and D only

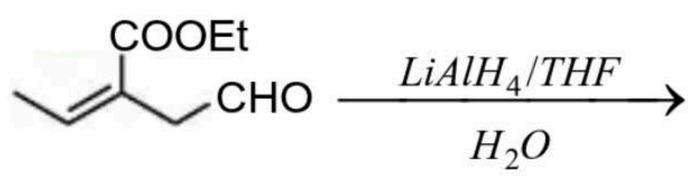
48. Match List I with List II.

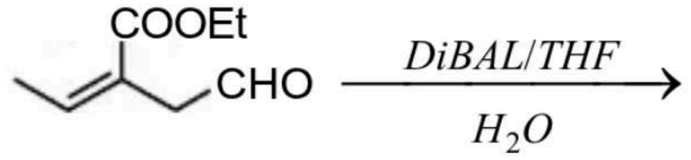
List I

(Reaction)

A. 

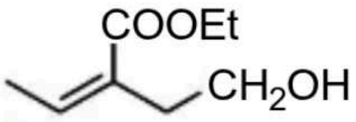
B. 

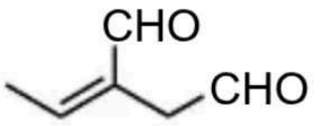
C. 

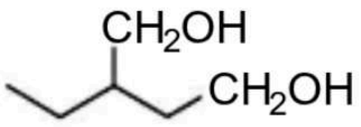
D. 

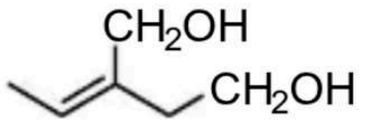
List II

(Product)

I. 

II. 

III. 

IV. 

Choose the correct answer from the options given below.

(1) A-III B-IV C-II D-I

(2) A-III B-I C-IV D-II

(3) A-II B-I C-IV D-III

(4) A-III B-IV C-I D-II

49. Given below are two statements.

Statement I : Heisenberg's uncertainty principle is significant for macroscopic particles.

Statement II : The product of uncertainties of both position and momentum for macroscopic particles is negligible.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

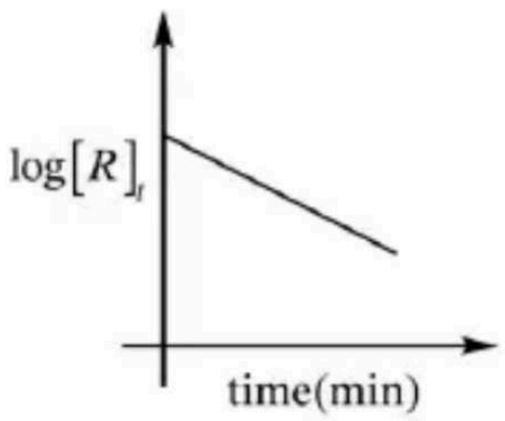
(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

50. Slope of the following graph is $-0.02\ min^{-1}$

Half-life of the reaction is ($[R]_t$ = amount remaining after time 't')



(1) 15 min (2) 30 min

(3) 20 min (4) 40 min

51. MX_2 is a strong electrolyte. Molar conductivity of 0.36 M MX_2 and 1.0 M MX_2 is 360 mho cm^2 mol^{-1} and 300 mho cm^2 mol^{-1}, respectively. Limiting molar conductivity of MX_2 at the same temperature is __ mho cm^2 mol^{-1}.

(1) 450 (2) 540

(3) 480 (4) 600

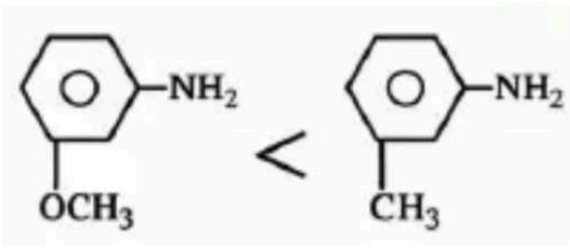
52. CFSE of $K_2[NiCl_4]$ is

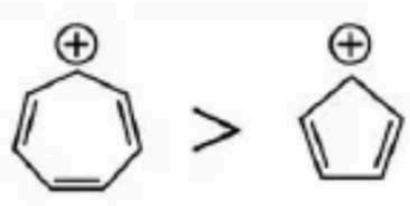
(1) $-0.8\ \Delta_t$ (2) $-1.2\ \Delta_t$

(3) zero (4) $-1.6\ \Delta_t$

53. Which of the following is an INCORRECT combination?

(1) methoxide > Ethoxide.....stability

(2)  ----basic strength

(3)  ------stability

(4) $C_6H_5-CH_2-COOH >$ C_6H_5COOH

------acidic strength

54. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Ethyl chloride reacts with ethanolic KCN to give Ethane nitrile.

Reason (R) : The bond enthalpy of a C–C single bond is greater than that of a C–N single bond.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

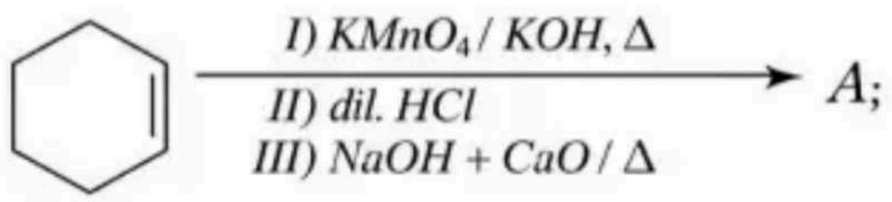
(4) (A) is false, but (R) is true.

55. Match List I with List II.

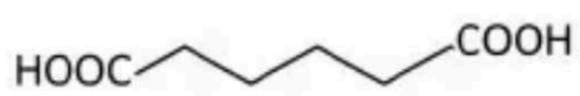
List I (Trend)	List II (Property)
A. $N < P < As < Bi < Sb$	I. Electro-negativity
B. $N > P > As > Sb = Bi$	II. Ionization potential
C. $N < P < Bi < Sb < As$	III. Melting point
D. $N > P > As > Sb > Bi$	IV. Boiling point

Choose the correct answer from the options given below.

(1) A-II B-I C-IV D-III
(2) A-III B-I C-IV D-II
(3) A-IV B-I C-III D-II
(4) A-IV B-II C-III D-I

56. 

Compound 'A' is

(1) 

(2) 

(3) 

(4) 

57. $CH_{4(g)} \rightarrow C_{(g)} + 4H_{(g)}$,
$\Delta H = +398 K.Cal / mole$

Which of the following is INCORRECT about the above reaction?

(1) ΔH is called enthalpy of atomisation.
(2) B.E of C-H bond is 398 K.Cal/mole.
(3) Average B.E of C-H bond is 99.5 K.Cal/mole.
(4) Energy required to break four moles of C-H bonds in CH_4 is 398 K.Cal.

58. $5A + 3B \longrightarrow 2C + 4D$. Four moles of 'A' and two moles of 'B' are allowed to react with each other. The correct statement is

(1) 1.6 moles of 'C' is formed.
(2) 2.66 moles of 'D' is formed.
(3) 'B' is the excess reagent.
(4) 0.33 moles of 'A' is left.

59. Wrong match is

(1) C_2H_5OH ----- positive iodoform test
(2) $(CH_3)_3COH$----Immediate turbidity in Lucas test
(3) C_6H_5OH....Violet colour with neutral $FeCl_3$
(4) $(CH_3)_2CHOH$.....on oxidation gives propanal

60. The freezing point of a solution containing 'n' moles of glucose in 200 grams of water is 272.7 K. When the same solution is cooled to –0.4°C, then the mass of water separated as ice will be

(1) 80g (2) 150 g
(3) 100 g (4) 50 g

61. A 0.02 M $BaCl_2$ solution is isotonic with 0.05 M glucose solution. The percentage ionization of $BaCl_2$ is

(1) 90% (2) 40%
(3) 75% (4) 60%

62. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : PbI_4 is stable and exists in nature.

Reason (R) : SnF_4 and PbF_4 are ionic in nature.

In light of the above statements, choose the correct answer from the options given below.

(1) (A) is false, but (R) is true.
(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are true, and (R) is the correct explanation of (A).

63. Which of the following is an INCORRECT combination about the given titrations?

(1) $KMnO_4$ *vs* $H_2C_2O_4$ in acidic medium---- $KMnO_4$ is the indicator

(2) $KMnO_4$ vs oxalic acid in acidic medium -----carried out at 333 K

(3) *NaOH vs* $H_2C_2O_4$ ----methyl red is the indicator

(4) *HCl vs* Na_2CO_3 ----methyl orange is the indicator

64. Given below are two statements.

Statement I : Bond angle in $SO_2 > NH_3 > H_2O > H_2S$.

Statement II : Hybridisation of central atom in SO_2 is 'sp'.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Statement I is correct, but statement II is incorrect.

65. The percentage of Iron in Misch metal is

(1) 5 (2) 0.5

(3) 1 (4) 3

66. Wrong match is

(1) $N(CH_2CH_2NH_2)_3$----Tetradentate ligand

(2) (en)----------didentate ligand

(3) $C_2O_4^{-2}$ .....Tridentate ligand

(4) SCN^-...ambidentate ligand

67. Identify the correct statement.

(1) Melting point of Ga > Tl.

(2) SeF_4 is a solid.

(3) In XeF_2, electron pairs are arranged as square pyramidal.

(4) Bond enthalpy $F_2 > I_2$

68. The reaction of sulphuric acid with sodium hydroxide is an example of

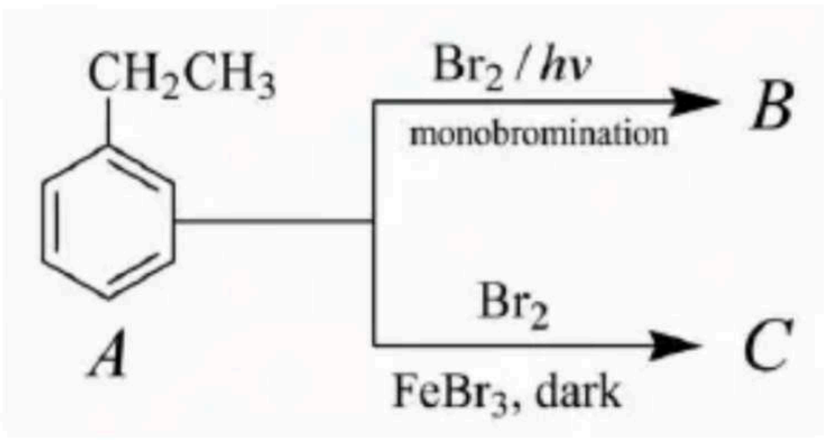
(1) comproportionation

(2) disproportionation

(3) non-redox reaction

(4) redox reaction

69.



Reactivity of (B) and (C) towards aq. NaOH through S_N1 mechanism is

(1) B = C (2) B ≈ C

(3) C > B (4) B > C

70. INCORRECT statements among the following are

A. The difference in the boiling point of chloroform and aniline is more than 50°C.

B. The boiling point of aniline is less than that of water.

C. In thin-layer chromatography, the retardation factor is least for the most adsorbed component.

D. Acetaldehyde is soluble in water, but insoluble in diethyl ether.

(1) B and D (2) A and C

(3) B, C, and D (4) C and D

71. Identify the INCORRECT statement.

(1) Formal charge on central nitrogen atom in N = N = O is '+1'.

(2) The number of pi bonds in the C_2 molecule are the same as in the N_2 molecule.

(3) Dipole moment of CCl_4 is less than CH_3Cl.

(4) An ionic halide with maximum ionic character will always have maximum lattice enthalpy.

72. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Solubility of $Zn(OH)_2$ is less in NaOH than in water.

Reason (R) : $Zn(OH)_2$ is not amphoteric in nature.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

73. The activation energy of a reaction depends on the

(1) temperature

(2) pressure

(3) concentration of reactants

(4) catalyst

74. Glucose $\rightarrow$ Gluconic acid. This conversion cannot be carried out using

(1) Concentrated nitric acid

(2) Tollen's reagent

(3) Fehling's reagent

(4) Br_2 water

75. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : The boiling point of n-pentane is less than the boiling point of 2-methyl Butane.

Reason (R) : Among the isomeric alkanes, as the branching increases, Vanderwal's forces of attraction decrease.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

76. Identify the INCORRECT statement.

(1) EO_2 and EO_3 type oxides (E = S, Se, Te) are acidic.

(2) The 17th group hydride with the highest melting point is HF.

(3) O_2PtF_6 is a red-coloured compound.

(4) Boron exhibits allotropy.

77. Select the correct reaction.

(1) 3-Methyl but-1-ene $\xrightarrow[Zn+H_2O]{O_3}$ Butan-2-one+Methanal

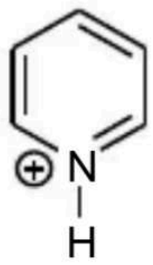
(2) But-2-ene $\xrightarrow[273K]{KMnO_4/OH^-}$ Acetic acid

(3) N-potassium phthalimide $\xrightarrow[III)H^+]{I)C_6H_5Cl \; II)aq.NaOH}$ Aniline

(4) Acetaldehyde $\xrightarrow[\Delta]{dil.NaOH}$ But-2-enal

78. $RCH_2OH \xrightarrow{X} R-COOH.$

'X' cannot be

(1)  CrO_3Cl^-

(2) $Conc.HNO_3$

(3) $KMnO_4 / H^+$

(4) $CrO_3 + H_2SO_4$

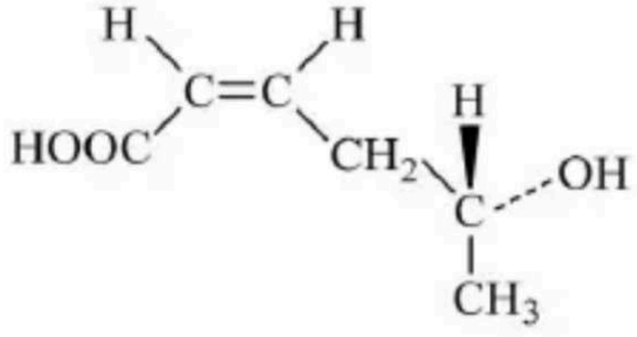
79. The number of electrons with

$n = 4, l = 2, m = 0, S = +\frac{1}{2}$ in 'Th' (Z=90)

is

(1) 1 (2) 3

(3) 6 (4) 4

80. 

The above molecule can show

(1) both geometrical and optical isomerism

(2) geometrical but not optical

(3) optical but not geometrical

(4) Neither geometrical nor optical

81. The general formula of an amino acid is

$H_2N - \underset{\displaystyle R}{\underset{|}{CH}} - COOH$

The correct match of 'R' and amino acid is

	List I (–R)		List II (Amino acid)
A.	$-\underset{\displaystyle CH_3}{\underset{\displaystyle \vert}{CH}} - CH_2 - CH_3$	I.	Tryptophan
B.	$-CH_2COOH$	II.	Aspartic acid
C.	$-CH_2 - \underset{\displaystyle CH_3}{\underset{\displaystyle \vert}{CH}} - CH_3$	III.	Leucine
D.	$-\underset{\displaystyle OH}{\underset{\displaystyle \vert}{CH}} - CH_3$	IV.	Isoleucine
		V.	Glutamic acid
		VI.	Threonine

Choose the correct answer from the options given below.

(1) A-III B-V C-IV D-VI

(2) A-IV B-V C-III D-I

(3) A-III B-II C-IV D-I

(4) A-IV B-II C-III D-VI

82. Which one of the following cannot decolourise either aqueous bromine or bromine in carbon tetrachloride?

(1) Benzene (2) Phenol

(3) Aniline (4) Vinyl benzene

83. $E^0_{A^{2+}/A} = 0.81V; E^0_{B^{2+}/B} = 0.41\ V$

If the reaction $A^{2+} + B \uparrow B^{2+} + A$ is at equilibrium, $\log K = \frac{4x}{3}$, then the value of 'x' is nearly

(1) 10 (2) 20

(3) 40 (4) 5

84. The correct statements among the following are

A. Lanthanum can acquire ccp, bcc, and hcp structures.

B. Melting point of Ag > Au > Cu.

C. V_2O_4 is acidic in nature.

D. In a faint alkaline medium, $KMnO_4$ oxidises thiosulphate ion to sulphate ion.

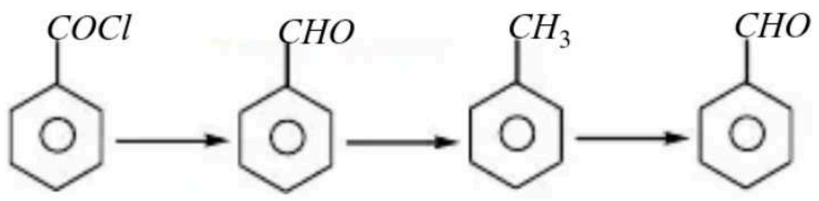
(1) A, C, and D only (2) A and D only

(3) B and D only (4) B and C only

85. $P_4 \xrightarrow{OH^-} PH_3 + H_2PO_2^-$. The number of OH^- ions in the net stoichiometric equation is

(1) 5 (2) 3

(3) 8 (4) 7

86. 

The above sequence consists of three named reactions. Identify the named reaction that is NOT involved in this sequence.

(1) Wolff–Kishner reduction

(2) Étard reaction

(3) Rosenmund's reduction

(4) Stephen's reaction

87. The incorrectly matched pair is:

(1) Starch.....lyophilic sol

(2) $Al(OH)_3$.....lyophobic sol

(3) CuS .....black

(4) CrO_2Cl_2 .....Green

88. Match List I with List II.

	List I (Atomic number)		**List II** (Element)
A.	99	I.	Cm
B.	107	II.	Mt
C.	109	III.	Es
D.	96	IV.	Bh

Choose the correct answer from the options given below.

(1) A-III B-IV C-II D-I

(2) A-I B-II C-III D-IV

(3) A-II B-III C-I D-IV

(4) A-III B-II C-IV D-I

89. At 298 K, pH of a solution is 6. If the concentration of hydronium ion is decreased by 100 times, then the pH of the resulting solution is

(1) 7.04

(2) 6.96

(3) 8

(4) 8.2

90. $2XY \Leftrightarrow X_2 + Y_2$. If $[X_2]_{eq} = 3\times10^{-4} M$; $[Y_2]_{eq} = 2.12\times10^{-4} M$ and $[XY]_{eq} = 1.8\times10^{-4} M$, then the degree of decomposition of XY, if we start with 0.5 moles of 'XY' initially, will be nearly

(1) 0.810

(2) 0.549

(3) 0.738

(4) 0.648

91. Study the following statements.

A. All organisms, including those in the plant and animal kingdoms, have species as the lowest category.

B. Some of the bacteria are autotrophic i.e., they synthesise their own food from sugars.

C. In slime moulds, during unfavourable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips.

D. During sexual reproduction, the fungi form fruiting bodies in which reduction division occurs leading to formation of haploid spores.

Select the correct option for the given statements.

(1) A, B, C, and D are correct.

(2) B, C, and D are incorrect.

(3) A, B, and C are correct.

(4) Only B is incorrect.

92. Which of the following statements regarding hormonal functions during the human menstrual cycle is correct?

(1) Luteinising hormone directly stimulates the proliferation of the endometrium during the follicular phase.

(2) Progesterone secreted after ovulation maintains the endometrium for implantation and inhibits further ovulation.

(3) Follicle-stimulating hormone is primarily responsible for maintaining the corpus luteum during the luteal phase.

(4) Oestrogen levels decline sharply just before ovulation, triggering the LH surge.

93. Which of the following is NOT a part of the ground tissue system?

(1) Mesophyll

(2) Pericycle

(3) Medulla

(4) Cuticle

94. Which of the following statements are true for Mollusca but NOT for Echinodermata?

A. Presence of a mantle that secretes a calcareous shell.

B. Body exhibits bilateral symmetry in adult stage.

C. Presence of a radula for feeding.

D. Water vascular system is present for locomotion.

E. Feather-like gills perform both respiration and excretion.

(1) A, B, and C only

(2) A, C, and E only

(3) B, C, and D only

(4) A, B, C, and E only

95. Given below are two statements.

Statement I : A coenzyme or metal ion that is tightly bound to an apoenzyme is called a prosthetic group.

Statement II : A complete catalytically active enzyme with its bound prosthetic group is called holoenzyme.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

96. Which of the following represents a limitation or drawback of Assisted Reproductive Technologies?

(1) It is universally affordable and easily accessible.

(2) It may involve ethical and legal concerns.

(3) It completely replaces natural fertilisation in humans.

(4) It eliminates the need for hormonal regulation.

97. Which of the following flowers ensure assured seed set even in the absence of pollinators?

(1) Chasmogamous flowers

(2) Cleistogamous flowers

(3) Flowers that open at the time of pollination

(4) Flowers that open early

98. Match List I with List II.

	List I		List II
A.	Ciliated epithelium	I.	Movement of mucus/particles
B.	Hepatic caeca of cockroach	II.	Secretion of digestive enzymes
C.	Dense regular connective tissue	III.	Tendons with parallel collagen fibres
D.	Fat bodies of frog	IV.	Storage of lipids

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-II B-I C-III D-IV

(3) A-I B-III C-II D-IV

(4) A-I B-II C-IV D-III

99. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : For Bt plants, the choice of genes (cry genes) depends on targeted pests.

Reason (R) : Bt toxins are insect group specific.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

100. Which of the following statements regarding sex determination is INCORRECT?

(1) In birds, the female is heterogametic (ZW) and the male is homogametic (ZZ).

(2) In humans, males produce two types of gametes with respect to sex chromosomes.

(3) Grasshoppers exhibit XO type of sex determination.

(4) In honeybees, fertilised eggs develop into males and unfertilised eggs into females.

101. Which of the following enzymes is NOT required during fermentation in microbes?

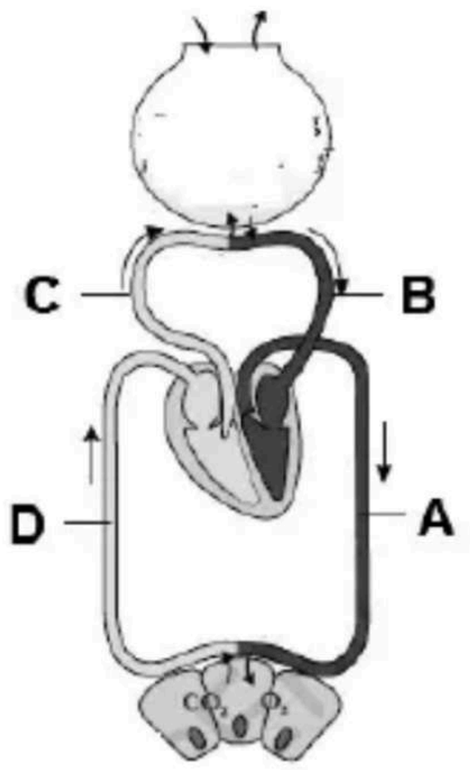
(1) Alcohol dehydrogenase

(2) Pyruvate dehydrogenase

(3) Lactic acid dehydrogenase

(4) Pyruvate decarboxylase

102. The given figure shows a schematic representation of blood circulation in humans with labels A to D. Select the option that correctly identifies a label along with its function.



(1) A - Systemic vein – carries oxygenated blood pO_2 = 95mm Hg

(2) B - Pulmonary vein - carries oxygenated blood pCO_2 = 45mm Hg

(3) C - Pulmonary artery - carries deoxygenated blood, pO_2 = 40mm Hg

(4) D - Systemic artery - carries deoxygenated blood pCO_2 = 45mm Hg

103. Match List I with List II.

	List I		List II
A.	Repetitive DNA	I.	Largest number of genes
B.	Coding DNA	II.	Least number of genes
C.	Chromosome 1	III.	Less than 2% of genome
D.	Chromosome Y	IV.	Major portion of human genome

Choose the correct answer from the options given below.

(1) A-IV B-III C-I D-II

(2) A-III B-IV C-I D-II

(3) A-IV B-III C-II D-I

(4) A-II B-III C-I D-IV

104. Match List I with List II.

	List I		List II
A.	*i - gene*	I.	transacetylase
B.	*lac-z*	II.	entry of β-galactosides
C.	*lac-a*	III.	hydrolysis of lactose
D.	*lac-y*	IV.	synthesis of repressor

Choose the correct answer from the options given below.

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-III, C-II, D-I

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-IV, C-I, D-II

105. Which of the following best describes lymph?

(1) A fluid rich in RBCs, platelets, and plasma proteins.

(2) A colourless fluid lacking RBCs, with fewer proteins than plasma.

(3) A fluid identical to blood plasma but without WBCs.

(4) A viscous fluid rich in haemoglobin and clotting factors.

106. Select the INCORRECT statement for maize plant.

(1) Production of C_3 acid in bundle sheath cells.

(2) Formation of PEP in bundle-sheath cells.

(3) Lack of photorespiration.

(4) Occurrence of C_3 pathway in bundle sheath cells.

107. Study the following statements.

A. Epicotyl of monocot embryo has a shoot apex and a few leaf primordia enclosed in a hollow foliar structure, the coleoptile.

B. Mendel investigated characters in garden pea plants that were manifested as two opposite traits.

C. When a phosphate group is linked to OH of $5'C$ of a nucleoside through phosphoester linkage, a corresponding nucleotide is formed.

D. The beads on a string structure in chromatin are packaged to form chromatin fibres that are further coiled and condensed at metaphase stage of cell division to form chromosomes.

Select the combination of correct statements.

(1) A, B, and C only

(2) A and B only

(3) A and C only

(4) A, B, C, and D

108. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : After a reduction in industrial pollution, the frequency of light-coloured moths increased in certain areas.

Reason (R) : Reappearance of lichens on tree trunks improved camouflage for light-coloured moths, thereby reducing predation.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

109. Arrange the following steps in the mechanism of ADH action in a correct sequence.

A. Increase in water reabsorption in distal tubule

B. Release of ADH from neurohypophysis

C. Activation of osmoreceptors

D. Loss of body fluid / increase in ionic concentration

E. Prevention of diuresis

(1) D → C → B → A → E

(2) C → D → B → A → E

(3) D → B → C → A → E

(4) D → C → A → B → E

110. During a seasonal outbreak, a patient presents with fever, mild rash, and joint pain. Laboratory findings show a moderate decrease in platelet count. After the fever subsides, the patient continues to experience marked joint stiffness that interferes with daily activities.

Which of the following diseases is most likely responsible?

(1) Malaria

(2) Chikungunya

(3) Typhoid

(4) Ascariasis

111. Which regions of the primary dicot stem are made up of simple mechanical tissues?

(1) Hypodermis and pith

(2) Endodermis and pericycle

(3) Hypodermis and pericycle

(4) Epidermis and endodermis

112. Given below are two statements.

Statement I : In replication unit of the DNA, one strand acts as a template while another acts as a coding strand.

Statement II : Nucleic acid polymerases enzymes can be classified on the basis of chemical nature of the template and the nature of nucleic acids synthesised from it.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

113. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Contraction of a skeletal muscle fibre requires calcium ions and ATP.

Reason (R) : Calcium ions expose the myosin-binding sites on actin, while ATP is required for cross-bridge cycling and for the detachment of myosin heads from actin.

In light of the above statements, choose the correct answer from the options given below.

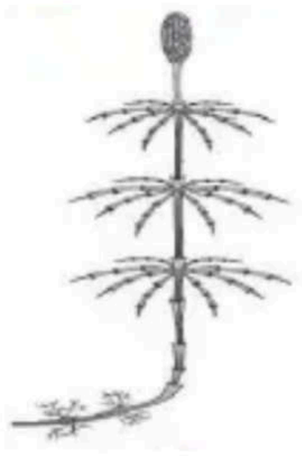
(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

114. Select the correct statement for the given organism.



(1) Heterosporous plant

(2) Amphibian of plant kingdom

(3) It produces independent dioecious male and female gametophytes.

(4) Tracheophyte without seed

115. A patient is suspected to have a viral infection, but the pathogen load in the blood is very low and cannot be detected by conventional methods. The doctor recommends a technique to amplify the genetic material of the pathogen.

Which of the following techniques is most appropriate for this purpose?

(1) ELISA (2) PCR

(3) MRI (4) CAT

116. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : In *Drosophila,* genes of body colour and eye colour did not segregate independently of each other, and the F_2 ratio deviated very significantly from Mendelian ratios.

Reason (R) : In *Drosophila,* genes of body colour and eye colour are showing physical association.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

117. Given below are two statements.

Statement I : The cerebellum coordinates voluntary muscular movements and maintains posture and balance.

Statement II : The cerebellum has a highly folded cortex, which increases its surface area.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

118. Lecithin is a

(1) phospholipid (2) glycolipid

(3) simple lipid (4) monoglyceride

119. Which of the following conditions is most likely to result in logistic population growth?

(1) Unlimited food and absence of predators

(2) Constant immigration with no emigration

(3) Limited resources and intraspecific competition

(4) High birth rate with zero death rate

120. Which of the following are correct regarding the cyclic photophosphorylation of photosynthesis?

A. It occurs on stroma thylakoids.

B. Photolysis of H_2O takes place.

C. Excited electron is cycled back to PS I.

D. Electrons are transported to NADP.

E. It involves only PS I.

(1) A, C, and E only

(2) A, B, and C only

(3) C, D, and E only

(4) B, D, and E only

121. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Hypersecretion of glucocorticoids may lead to hyperglycemia.

Reason (R) : Glucocorticoids inhibit gluconeogenesis in liver cells.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

122. Arrangement of a, b, c, d genes on a chromosome with map units are

$a-d=4$, $b-c=1$, $a-b=7$, $c-d=2$, $a-c=6$

What is the correct sequence of genes?

(1) a – d – c – b (2) a – c – d – b

(3) d – c – b – a (4) a – b – c – d

123. The following is a grazing food chain:

Phytoplankton → Zooplankton → Fish → Bird

In the above chain, the bird is __A__. If the net secondary productivity at the fish level is 20 kJ, then GPP of phytoplankton is approximately __B__ (Assume 10% law, and R in producers = 20% of GPP).

(1) A – Secondary consumer, B – 2500 kJ

(2) A – Tertiary consumer, B – 2500 kJ

(3) A – Tertiary consumer, B – 2000 kJ

(4) A – Top carnivore, B – 3000 kJ

124. Match List I with List II.

	List I	**List II**
A.	Pollination service	I. Broadly utilitarian value
B.	Food crops	II. Narrowly utilitarian value
C.	Sacred groves	III. *In situ* conservation
D.	Cryopreservation	IV. *Ex situ* conservation

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-II B-I C-III D-IV

(3) A-I B-III C-II D-IV

(4) A-IV B-III C-II D-I

Sri Chaitanya Educational Institutions | Infinity Learn

125. Given below are two statements.

Statement I : In photophosphorylation, it is the light energy that is utilised for production of proton gradient, required for phosphorylation.

Statement II : The number of ATP molecules synthesised during respiration depends on the nature of the electron donor.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

126. Which of the following statements correctly differentiates between the male and female reproductive systems?

(1) The vas deferens serves as the primary site for sperm maturation, whereas the fallopian tube serves as the primary site for ovum maturation.

(2) Seminiferous tubules are the site of gamete formation, whereas ovaries lack any comparable organised structures for gamete development.

(3) The male reproductive system includes accessory glands, whereas the female system does not possess accessory glands.

(4) In males, the urethra serves as a common duct for both the urinary and reproductive systems, whereas in females it functions exclusively as a urinary duct.

127. In which of the following aspects, does the dorsiventral leaves NOT differ from isobilateral leaves?

A. Orientation on stem

B. Differentiation of mesophyll

C. Distribution of stomata

D. Type of vascular bundle

E. Position of xylem and phloem

F. Venation type

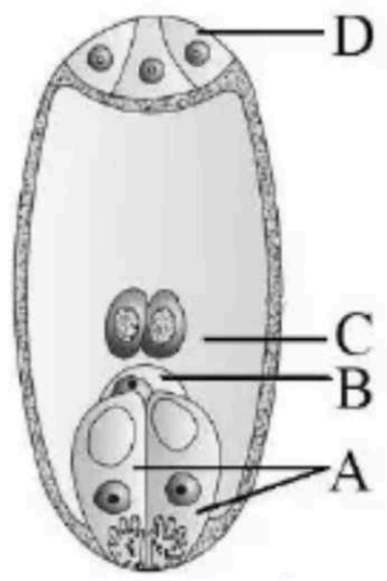
(1) A, B, and C only

(2) D, E, and F only

(3) A, D, E, and F only

(4) D and E only

128. Select the INCORRECT statement for the given structure.



(1) *A* – helps to guide the pollen tube

(2) *B* – converts into an embryo after fertilisation

(3) *C* – converts into cotyledons in the mature seed

(4) The diagram represents a female gametophyte of flowering plants

129. Which of the following features is common to Pisces, Amphibia, Reptilia, and Aves?

(1) Three-chambered heart

(2) Nucleated erythrocytes

(3) Internal fertilisation in all members

(4) Pulmonary respiration in all members

130. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Amniocentesis is banned for determining the sex of the foetus in India.

Reason (R) : It can be misused for sex-selective abortion (female foeticide).

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

131. Match List I with List II.

	List I		**List II**
A.	OAA	I.	4C
B.	Phosphoglycolic acid	II.	3C
C.	Lactic acid	III.	2C
D.	Isocitric acid	IV.	6C

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-I B-III C-IV D-II

(3) A-I B-III C-II D-IV

(4) A-IV B-III C-II D-I

132. Which of the following features is NOT common to cockroaches and frogs?

(1) Bilateral symmetry

(2) Triploblastic organisation

(3) Open circulatory system

(4) Complete digestive system

133. Given below are two statements.

Statement I : The BOD test measures the rate of uptake of oxygen by microorganisms in a sample of water.

Statement II : Baculoviruses are excellent candidates for species-specific, narrow-spectrum insecticidal applications.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

134. Which of the following statements are correct?

A. Colour blindness is a sex-linked recessive disorder in which the affected individual fails to distinguish between red and green colours.

B. Sickle cell anaemia is caused by substitution of glutamic acid by valine at the sixth position of β-globin chain.

C. Down syndrome occurs due to trisomy of chromosome 21 as a result of non-disjunction.

D. Individuals exhibiting Klinefelter syndrome possess 44 autosomes and only two sex (X) chromosomes.

E. Phenylketonuria is due to the deficiency of an enzyme that converts phenylalanine into tyrosine.

(1) A, B, C, and E

(2) A, C, and D only

(3) B, C, D, and E

(4) A, B, C, D, and E

135. Homologous chromosomes separate while sister chromatids remain associated at their centromeres in

(1) anaphase II (2) metaphase II

(3) anaphase I (4) metaphase I

136. Which of the following statements are INCORRECT?

A. Tidal volume is the only component common to both inspiratory capacity and vital capacity.

B. Inspiratory reserve volume is a component common to both inspiratory capacity and total lung capacity.

C. Residual volume is the only component common to both functional residual capacity and total lung capacity.

D. Expiratory reserve volume is the only component common to both vital capacity and functional residual capacity.

E. Inspiratory reserve volume is the only component common to both vital capacity and total lung capacity

(1) A, C, and E (2) A and E only

(3) C and D only (4) B, D, and E

137. Which of the following is NOT a cause of H^+ gradient development in chloroplast?

(1) Splitting of H_2O

(2) Pumping of H^+ from the stroma to the lumen.

(3) Oxidation of $NADH + H^+$ and $FADH_2$

(4) Activity of FNR

138. Match List I with List II.

List I	List II
A. Abscisic acid	I. Delays leaf senescence
B. Ethylene	II. Inhibits seed germination
C. Cytokinin	III. Herbicide
D. Auxin	IV. Breaks seed and bud dormancy

Choose the correct answer from the options given below.

(1) A-II, B-IV, C-I, D-III

(2) A-I, B-II, C-III, D-IV

(3) A-II, B-III, C-IV, D-I

(4) A-II, B-I, C-III, D-IV

139. Which of the following statements are correct?

A. AV valves open only when ventricular pressure exceeds atrial pressure.

B. End-systolic volume refers to the volume of blood left in each ventricle at the end of ventricular systole.

C. Atrial systole overlaps with ventricular systole.

D. Atrial systole contributes only a fraction to ventricular filling under normal conditions.

E. During atrial diastole, atria actively pump blood into ventricles.

(1) B and D only (2) A and C only

(3) B, C, and D (4) A, D, and E

140. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : The maximum concentration of solutes in the medullary interstitium is achieved at the inner medulla.

Reason (R) : This is mainly due to the active transport of NaCl from the descending limb of the loop of Henle.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

141. Which of the following traits is exhibited in both homozygous and heterozygous states in the pea plant?

(1) Terminal flower (2) Green seed

(3) White flower (4) Green pod

142. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : Accessory pigments enhance the rate of photosynthesis in higher plants.

Reason (R) : Accessory pigments absorb wavelengths of light that are not efficiently absorbed by chlorophyll.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

143. Which of the following statements regarding the effects of drugs is INCORRECT?

(1) Morphine acts as a sedative and painkiller.

(2) Barbiturates act as depressants and induce sleep.

(3) Tranquilisers relieve anxiety and reduce mental tension.

(4) Cocaine decreases heart rate and reduces blood pressure.

144. A person is unable to bring the thumb across the palm to touch the fingers, affecting grip and precision movements. This is most likely due to dysfunction of which joint?

(1) Gliding joint

(2) Saddle joint

(3) Ball and socket joint

(4) Pivot joint

145. Identify the mismatch regarding the enzyme and its function.

(1) Bacterial cell wall dissolution $\rightarrow$ ligase

(2) Fungal cell wall dissolution $\rightarrow$ chitinase

(3) Plant cell wall dissolution $\rightarrow$ cellulase

(4) PCR $\rightarrow$ Taq DNA polymerase

146. Given below are two statements.

Statement I : Transgenic animals are mainly used for increasing agricultural crop yield.

Statement II : Transgenic animals can be used to produce therapeutic proteins such as human alpha-1 antitrypsin.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

147. Given below are two statements.

Statement I : Embryo develops at the micropylar end of the embryo sac.

Statement II : The central cell after triple fusion becomes the primary endosperm cell.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

148. Match List I with List II.

	List I	**List II**
A.	Predation	I. One species benefits without affecting the other
B.	Competition	II. Detrimental to both species due to shared limited resources
C.	Mutualism	III. One organism kills another for food
D.	Commensalism	IV. Both species show increased fitness and survival

Choose the correct answer from the options given below.

(1) A-III B-II C-IV D-I

(2) A-II B-III C-IV D-I

(3) A-III B-IV C-II D-I

(4) A-IV B-II C-III D-I

149. Which of the following is NOT concerned with gibberellins?

(1) Bolting

(2) Seed germination

(3) Delay of senescence of fruits

(4) Thinning of flowers and fruits

150. Given below are two statements.

Statement I : Hypothalamic hormones regulating anterior pituitary are neurohormones secreted by neurosecretory cells.

Statement II : The pituitary is anatomically connected to the hypothalamus by a stalk.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

151. Match List I with List II.

	List I	**List II**
A.	SER	I. Synthesis of lipids
B.	Lysosome	II. Mechanical support
C.	Cytoskeleton	III. Spindle apparatus
D.	Centriole	IV. Digestion of carbohydrates

Choose the correct answer from the options given below.

(1) A-I, B-II, C-III, D-IV

(2) A-I, B-IV, C-II, D-III

(3) A-I, B-IV, C-III, D-II

(4) A-IV, B-III, C-II, D-I

Sri Chaitanya Educational Institutions | Infinity Learn

152. In a tree ecosystem, a single large tree supports a large number of insects, which in turn support a smaller number of small birds, which further support an even smaller number of large birds. The pyramid of numbers in this case is

(1) upright (2) inverted

(3) spindle-shaped (4) square-shaped

153. Choose the correct option.

A. Oxidation of glyceraldehyde-3-phosphate

B. Oxidation of malic acid

C. Decarboxylation of pyruvic acid

D. Oxidative decarboxylation of isocitric acid

Arrange the following events of aerobic respiration for one molecule of glucose, in chronological manner.

(1) A → B → C → D

(2) D → C → B → A

(3) A → C → B → D

(4) A → C → D → B

154. The disappearance of a specific pollinator insect leading to the extinction of a plant species that depends exclusively on it is an example of

(1) habitat loss

(2) overexploitation

(3) alien species invasion

(4) coextinction

155. Study the following statements.

A. In tracheids, the inner layers of the cell walls have thickenings which vary in form.

B. The vascular bundles of the grass stem contain water-storing cavities.

C. The structure and functions of tissue are also dependent on location.

D. In dicot stem, the parenchymatous cells, which lie between xylem and phloem, are called conjunctive tissue.

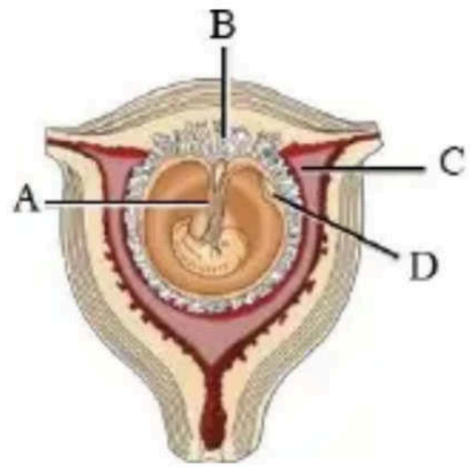
E. In dorsiventral leaves, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

How many statements are true?

(1) 4 (2) 3

(3) 2 (4) 5

156. Which of the following is the correct option regarding A to D from the given figure?



(1) *A* – Cavity of uterus, *B* – Placental villi, *C* – Umbilical cord, *D* – Yolk sac

(2) *A* – Umbilical cord, *B* – Amniotic villi, *C* – Cavity of uterus, *D* – Allantois

(3) *A* – Umbilical cord, *B* – Placental villi, *C* – Cavity of uterus, *D* – Yolk sac

(4) *A* – Cavity of uterus, *B* – Amniotic villi, *C* – Umbilical cord, *D* – Allantois

157. Given below are two statements. One is labelled Assertion (A), and the other is labelled Reason (R).

Assertion (A) : C_4 plants have greater productivity of biomass.

Reason (R) : C_4 plants lack a process called photorespiration.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

Sri Chaitanya Educational Institutions | Infinity Learn

158. Which of the following is INCORRECT regarding cellulose?

(1) Homopolymer

(2) Made up of glucose monomers

(3) Does not contain complex helices and hence cannot hold iodine

(4) Storehouse of energy in plant tissues

159. Which of the following sets of animals belongs to the group Tetrapoda?

(1) *Salmo*, *Scoliodon*, *Salpa*

(2) *Panthera*, *Pavo*, *Petromyzon*

(3) *Columba*, *Crocodylus*, *Canis*

(4) *Labeo*, *Latimeria*, *Lepidosteus*

160. Which of the following is NOT a function of intrauterine devices?

(1) Increasing phagocytosis of sperm within the uterus

(2) Suppressing sperm motility and fertilising capacity

(3) Preventing implantation by altering the uterine environment

(4) Promoting ovulation by stimulating gonadotropin release

161. State true (T) or false (F) for the following statements.

A. The technology of biogas production was developed in India due to the efforts of IARI and KVIC.

B. The composition of biogas produced by *Methanobacterium* is largely methane along with CO_2 and H_2.

C. In a biogas plant, the spent slurry is removed through an outlet used as fertiliser.

D. Human beings can digest the cellulose present in foods because of the presence of *Methanobacterium* in their gut.

(1) A – T, B – F, C – T, D – F

(2) A – F, B – T, C – F, D – T

(3) A – F, B – F, C – T, D – T

(4) A – T, B – T, C – T, D – F

162. Given below are two statements.

Statement I : The matrix of cartilage is mineralised with calcium salts, which impart rigidity.

Statement II : Osteocytes are present in spaces called lacunae in bone.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

163. In papaya, dioecious condition prevents

(1) autogamy, but not geitonogamy

(2) geitonogamy, but not autogamy

(3) autogamy and geitonogamy

(4) autogamy only

164. Which of the following pairs is correctly matched?

(1) Homologous organs – Same function, different origin

(2) Analogous organs – Same origin, different function

(3) Homologous organs – Same origin, different functions

(4) Analogous organs – Different origin, different functions

165. Given below are two statements.

Statement I : RNA was the first genetic material.

Statement II : DNA has evolved from RNA with chemical modifications.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but statement II is correct.

(4) Both statement I and statement II are incorrect.

166. Select the methods of transmission of AIDS.

A. Unprotected sexual contact with an infected person

B. Use of common toilets and swimming pools

C. Transplantation of infected organs

D. Breastfeeding by an infected mother

E. Sharing food with an infected person

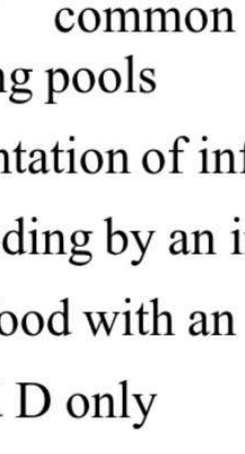
(1) A, C, and D only

(2) A, B, and C only

(3) A, C, D, and E only

(4) B, C, and D only

167. Choose the correct option with respect to the image given below.



(1) The above organism is a member of chrysophyte.

(2) The above organism is generally/mostly found in freshwater environments.

(3) The above organism contains cell wall that has stiff silica plates on the outer surface.

(4) It causes red tides.

168. Given below are two statements.

Statement I : Humification leads to the accumulation of a dark-coloured, amorphous substance called humus.

Statement II : Humus is highly resistant to microbial action and decomposes very slowly.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

169. Sclerenchyma cells are

(1) living, thin-walled and with large vacuoles

(2) living thick-walled and rich in cytoplasm

(3) dead, thin-walled and perform photosynthesis

(4) dead, thick-walled and provide mechanical support

170. Which of the following is a reason for greater biological diversity in the tropics?

(1) Long evolutionary time

(2) Frequent climatic disturbances

(3) Low primary productivity

(4) Repeated glaciation events

171. Which of the following statements holds true for oogenesis but NOT for spermatogenesis in humans?

A. Initiation of the process occurs during puberty.

B. The primary gametocyte undergoes prolonged arrest in prophase I.

C. Cytokinesis during meiotic divisions is unequal.

D. The process results in the formation of a single large functional gamete.

E. Meiosis II is completed prior to fertilisation.

Select the correct option.

(1) A, B, and C only

(2) A, B, and D only

(3) B, C, and D only

(4) A, C, and E only

172. In sickle cell anaemia, defective / altered mRNA has which genetic codon for amino acid valine?

(1) CTC (2) GAG

(3) CAC (4) GUG

173. Match List I with List II.

List I (Genus)		**List II (Characteristic)**
A.	*Ascaris*	I. Complete digestive system and no circulatory system
B.	*Nereis*	II. Closed circulatory system with nephridia for excretion
C.	*Periplaneta*	III. Jointed appendages with haemocoel functioning as body cavity
D.	*Balanoglossus*	IV. Presence of pharyngeal gill slits and stomochord

Choose the correct answer from the options given below.

(1) A-I B-II C-III D-IV

(2) A-II B-III C-I D-IV

(3) A-I B-III C-II D-IV

(4) A-III B-II C-I D-IV

174. Arrange the following events of translation in chronological manner.

A. Amino acylation of t-RNA (charging).

B. Encounter of small subunit with m-RNA.

C. Joining of initiator t-RNA.

D. Joining of charged t-RNA on 'A' site.

E. Peptide bond formation.

Choose the correct option.

(1) A → B → C → D → E

(2) A → C → B → D → E

(3) A → B → D → C → E

(4) A → E → D → C → B

175. Which of the following statements regarding the circulatory system of frogs is INCORRECT?

(1) Frogs possess a three-chambered heart consisting of two atria and one ventricle.

(2) Sinus venosus receives blood from major veins and opens into the right atrium.

(3) Hepatic portal system carries oxygenated blood from the liver directly to the heart.

(4) Conus arteriosus regulates the distribution of blood into the aortic arches.

176. Given below are two statements.

Statement I : Radial type of vascular bundles are present in anatomy of roots.

Statement II : Conjoint, collateral, and open types of vascular bundles are present in stems of monocots.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

177. Given below are two statements

Statement I : *Homo sapiens* originated in Africa and later migrated to other continents.

Statement II : *Homo erectus* was the first to develop advanced language similar to modern humans.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

178. Study the following and identify the INCORRECT statement/s.

A. Loose tissues are present in fungi.

B. Bacteria show the most extensive metabolic diversity.

C. Heterotrophic bacteria are less abundant in nature.

D. Pigments of euglenoids are identical to those present in higher plants.

(1) B, C, and D only

(2) C, and D only

(3) C only

(4) B and C only

179. The major limiting factor for photosynthesis in plants under natural conditions is

(1) carbon dioxide concentration

(2) oxygen concentration

(3) light intensity

(4) water availability

180. How many statements are correct from the following list?

A. Nucleus as a cell organelle was first described by Robert Brown as early as 1831.

B. The interphase nucleus (nucleus of a cell when it is not dividing) has highly extended and elaborate nucleoprotein fibres called chromatin.

C. The outer membrane of the nucleus usually remains continuous with the endoplasmic reticulum and bears ribosomes on it.

D. Some mature cells lack a nucleus, e.g., erythrocytes of many mammals and sieve tube cells of vascular plants.

E. The nucleoli are spherical structures present in the nucleoplasm of the nucleus.

(1) 4 (2) 5

(3) 2 (4) 1